

A Time-Discretized Model and Heuristic for Breast Cancer Radiotherapy

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October 15, 2023

2023 INFORMS Annual Meeting
Phoenix, Arizona

Outline

Introduction

Model Formulation

Heuristic Solution Method

Results

Conclusion

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What is Intensity Modulated Radiation Therapy (IMRT)?



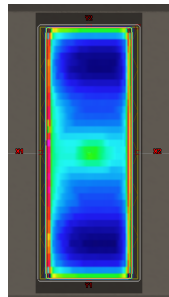
LINAC. Source: Top Doctors UK



MLC. Source: Varian Medical Systems

- ▶ IMRT is a cancer treatment method in which a high energy beam irradiates cancerous tissue

Sliding Window IMRT



Sliding leaf motion. Source: Vienna Health Association

Fluence map. Source: Vienna Health Association

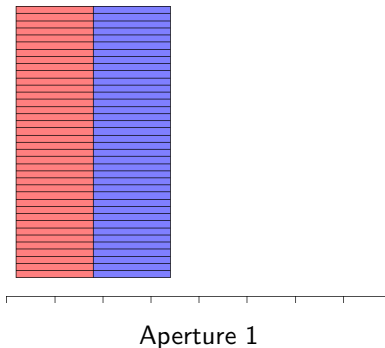
- ▶ A delivery modality in which the leaves of the MLC move unidirectionally across the beam-field
- ▶ Treatment planning is an iterative and manual process

Current Literature

- ▶ Brahme et al., 1982 first pioneered "inverse-planning" for rotational therapies.
- ▶ Convery and Rosenbloom, 1992 first proposed the sliding window method and showed that arbitrary dose patterns could be produced
- ▶ Papp and Unkelbach, 2014 modelled a sliding-window volumetric modulated arc therapy problem
- ▶ Craft and Halabi, 2018 adapted the work by Papp and Unkelbach to the regular sliding-window method
- ▶ Habib et al., 2022 sought to model the sliding-window IMRT model with the addition of several novel deliverability constraints

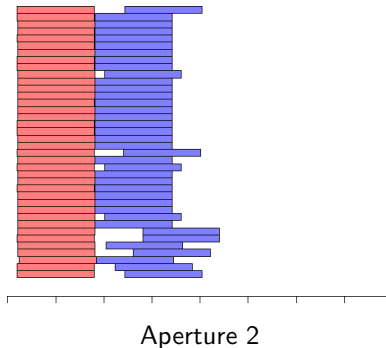
Motivation - Machine Input

- ▶ The sliding leaf motion is specified by a sequence of apertures
- ▶ An aperture specifies the position of the leaves at one instance in time
- ▶ Motion is interpolated linearly between apertures
- ▶ The LINAC can only take a limited number of apertures



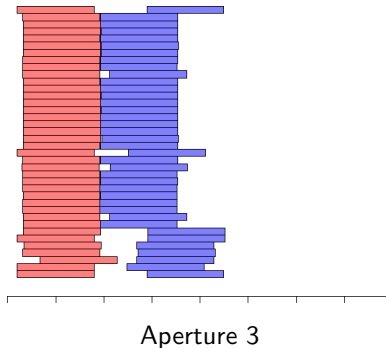
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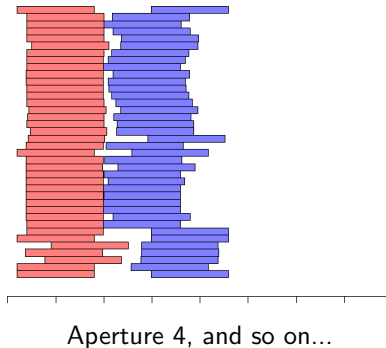
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Motivation - Recent Work

- ▶ Formulated the problem using a continuous time model
- ▶ The beamfield is divided into a grid of beamlets and for each beamlet boundary we get four values:
 - ▶ time the left leaf arrives and departs a particular boundary
 - ▶ time the right leaf arrives and departs a particular boundary

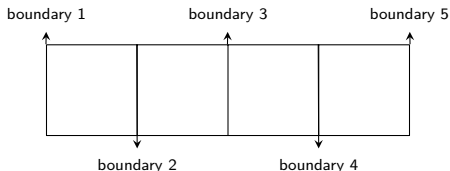


Figure: Beamlets boundaries of some row n

Problem Statement

- ▶ Solution produces very granular movement, which requires upwards of 3000 apertures to describe
- ▶ Solution format and model do not inherently account for limited apertures

The old models do not account for limited apertures put into the software.

Contribution

- ▶ **Modelling:** We propose a novel MINLP formulation of the problem, re-writing all required deliverability constraints under our new formulation
- ▶ **Heuristic:** We propose a heuristic approach to generate high-quality feasible solutions for the proposed MINLP model
- ▶ **Application:** Demonstrate that the proposed approach can generate clinically deliverable plans

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Aperture Definition

Variables:

- ▶ $l_{n,b}, r_{n,b} \in \{0, 1\}$
- ▶ $a_{n,b} = -1 + l_{n,b} + r_{n,b}$

No overlapping leaves:

- ▶ $l_{n,b} + r_{n,b} \geq 1$

Enforce unbroken leaves:

- ▶ $l_{n,b+1} \geq l_{n,b}$
- ▶ $r_{n,b} \geq r_{n,b+1}$

$l_{1,1} = 0$ $r_{1,1} = 1$ $a_{1,1} = 0$	$l_{1,2} = 1$ $r_{1,2} = 1$ $a_{1,2} = 1$	$l_{1,3} = 1$ $r_{1,3} = 0$ $a_{1,3} = 0$	$l_{1,4} = 1$ $r_{1,4} = 0$ $a_{1,4} = 0$
$l_{2,1} = 0$ $r_{2,1} = 1$ $a_{2,1} = 0$	$l_{2,2} = 1$ $r_{2,2} = 1$ $a_{2,2} = 1$	$l_{2,3} = 1$ $r_{2,3} = 1$ $a_{2,3} = 1$	$l_{2,4} = 1$ $r_{2,4} = 0$ $a_{2,4} = 0$
$l_{3,1} = 0$ $r_{3,1} = 1$ $a_{3,1} = 0$	$l_{3,2} = 0$ $r_{3,2} = 1$ $a_{3,2} = 0$	$l_{3,3} = 1$ $r_{3,3} = 0$ $a_{3,3} = 0$	$l_{3,4} = 1$ $r_{3,4} = 0$ $a_{3,4} = 0$

Figure: Dummy 3×4 beamlet grid

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$a_{1,1}^1 = 0$	$a_{1,2}^1 = 0$	$a_{1,3}^1 = 0$	$a_{1,4}^1 = 0$
$l_{2,1}^1 = 0$	$l_{2,2}^1 = 1$	$l_{2,3}^1 = 1$	$l_{2,4}^1 = 1$
$r_{2,1}^1 = 1$	$r_{2,2}^1 = 1$	$r_{2,3}^1 = 0$	$r_{2,4}^1 = 0$
$a_{2,1}^1 = 0$	$a_{2,2}^1 = 1$	$a_{2,3}^1 = 0$	$a_{2,4}^1 = 0$
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$a_{3,1}^1 = 0$	$a_{3,2}^1 = 0$	$a_{3,3}^1 = 0$	$a_{3,4}^1 = 0$



$l_{1,1}^2 = 0$	$l_{1,2}^2 = 1$	$l_{1,3}^2 = 1$	$l_{1,4}^2 = 1$
$r_{1,1}^2 = 1$	$r_{1,2}^2 = 1$	$r_{1,3}^2 = 0$	$r_{1,4}^2 = 0$
$a_{1,1}^2 = 0$	$a_{1,2}^2 = 1$	$a_{1,3}^2 = 0$	$a_{1,4}^2 = 0$
$l_{2,1}^2 = 0$	$l_{2,2}^2 = 0$	$l_{2,3}^2 = 1$	$l_{2,4}^2 = 1$
$r_{2,1}^2 = 1$	$r_{2,2}^2 = 1$	$r_{2,3}^2 = 1$	$r_{2,4}^2 = 0$
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Aperture 1 at t secondsAperture 2 at $2t$ secondsAperture 3 at $3t$ seconds

- There are a total of J apertures in the sequence, where J is a parameter determined beforehand

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$r_{1,1}^3 = 1$	$r_{1,2}^3 = 1$	$r_{1,3}^3 = 1$	$r_{1,4}^3 = 0$
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Unidirectional Motion & Velocity

Leaves must move left to right:

- ▶ $\mu_{n,b}^{j+1} \leq \mu_{n,b}^j$
- ▶ $r_{n,b}^{j+1} \geq r_{n,b}^j$

Distance travelled variable:

- ▶ $L_n^j = \sum_{b=1}^B \mu_{n,b}^j - \sum_{b=1}^B \mu_{n,b}^{j+1}$
- ▶ $R_n^j = \sum_{b=1}^B r_{n,b}^{j+1} - \sum_{b=1}^B r_{n,b}^j$

Restrict max distance travelled:

- ▶ $L_n^j \leq Y$
- ▶ $R_n^j \leq Y$

Aperture j

$l_{1,1}^1 = 0$	$l_{1,2}^1 = 1$	$l_{1,3}^1 = 1$	$l_{1,4}^1 = 1$	$l_{1,4}^1 = 1$
$r_{1,1}^1 = 1$	$r_{1,2}^1 = 1$	$r_{1,3}^1 = 1$	$r_{1,4}^1 = 0$	$r_{1,4}^1 = 0$
$a_{1,1}^1 = 0$	$a_{1,2}^1 = 1$	$a_{1,3}^1 = 1$	$a_{1,4}^1 = 0$	$a_{1,4}^1 = 0$

Aperture $j + 1$

$l_{1,1}^2 = 0$	$l_{1,2}^2 = 0$	$l_{1,3}^2 = 0$	$l_{1,4}^2 = 1$	$l_{1,4}^2 = 1$
$r_{1,1}^2 = 1$	$r_{1,2}^2 = 1$	$r_{1,3}^2 = 1$	$r_{1,4}^2 = 1$	$r_{1,4}^2 = 0$
$a_{1,1}^2 = 0$	$a_{1,2}^2 = 0$	$a_{1,3}^2 = 0$	$a_{1,4}^2 = 1$	$a_{1,4}^2 = 0$

$$\underbrace{L_n^j = 2} \quad \underbrace{R_n^j = 1}$$

Unidirectional Motion & Velocity

Leaves must move left to right:

$$\blacktriangleright \mu_{n,b}^{j+1} \leq \mu_{n,b}^j$$

$$\blacktriangleright r_{n,b}^{j+1} \geq r_{n,b}^j$$

Distance travelled variable:

$$\blacktriangleright L_n^j = \sum_{b=1}^B \mu_{n,b}^j - \sum_{b=1}^B \mu_{n,b}^{j+1}$$

$$\blacktriangleright R_n^j = \sum_{b=1}^B r_{n,b}^{j+1} - \sum_{b=1}^B r_{n,b}^j$$

Restrict max distance travelled:

$$\blacktriangleright L_n^j \leq Y$$

$$\blacktriangleright R_n^j \leq Y$$

Aperture j

$l_{1,1}^1 = 0$	$l_{1,2}^1 = 1$	$l_{1,3}^1 = 1$	$l_{1,4}^1 = 1$	$l_{1,4}^1 = 1$
$r_{1,1}^1 = 1$	$r_{1,2}^1 = 1$	$r_{1,3}^1 = 1$	$r_{1,4}^1 = 0$	$r_{1,4}^1 = 0$
$a_{1,1}^1 = 0$	$a_{1,2}^1 = 1$	$a_{1,3}^1 = 1$	$a_{1,4}^1 = 0$	$a_{1,4}^1 = 0$

Aperture $j + 1$

$l_{1,1}^2 = 0$	$l_{1,2}^2 = 0$	$l_{1,3}^2 = 0$	$l_{1,4}^2 = 1$	$l_{1,4}^2 = 1$
$r_{1,1}^2 = 1$	$r_{1,2}^2 = 1$	$r_{1,3}^2 = 1$	$r_{1,4}^2 = 1$	$r_{1,4}^2 = 0$
$a_{1,1}^2 = 0$	$a_{1,2}^2 = 0$	$a_{1,3}^2 = 0$	$a_{1,4}^2 = 1$	$a_{1,4}^2 = 0$

$$L_n^j = 2 \quad R_n^j = 1$$

Unidirectional Motion & Velocity

Leaves must move left to right:

$$\blacktriangleright \mu_{n,b}^{j+1} \leq \mu_{n,b}^j$$

$$\blacktriangleright r_{n,b}^{j+1} \geq r_{n,b}^j$$

Distance travelled variable:

$$\blacktriangleright L_n^j = \sum_{b=1}^B \mu_{n,b}^j - \sum_{b=1}^B \mu_{n,b}^{j+1}$$

$$\blacktriangleright R_n^j = \sum_{b=1}^B r_{n,b}^{j+1} - \sum_{b=1}^B r_{n,b}^j$$

Restrict max distance travelled:

$$\blacktriangleright L_n^j \leq Y$$

$$\blacktriangleright R_n^j \leq Y$$

Aperture j

$l_{1,1}^1 = 0$	$l_{1,2}^1 = 1$	$l_{1,3}^1 = 1$	$l_{1,4}^1 = 1$	$l_{1,4}^1 = 1$
$r_{1,1}^1 = 1$	$r_{1,2}^1 = 1$	$r_{1,3}^1 = 1$	$r_{1,4}^1 = 0$	$r_{1,4}^1 = 0$
$a_{1,1}^1 = 0$	$a_{1,2}^1 = 1$	$a_{1,3}^1 = 1$	$a_{1,4}^1 = 0$	$a_{1,4}^1 = 0$

Aperture $j + 1$

$l_{1,1}^2 = 0$	$l_{1,2}^2 = 0$	$l_{1,3}^2 = 0$	$l_{1,4}^2 = 1$	$l_{1,4}^2 = 1$
$r_{1,1}^2 = 1$	$r_{1,2}^2 = 1$	$r_{1,3}^2 = 1$	$r_{1,4}^2 = 1$	$r_{1,4}^2 = 0$
$a_{1,1}^2 = 0$	$a_{1,2}^2 = 0$	$a_{1,3}^2 = 0$	$a_{1,4}^2 = 1$	$a_{1,4}^2 = 0$

$$L_n^j = 2 \quad R_n^j = 1$$

Dose Requirements

Let T denote the set of tumour voxels and O denote the set of healthy tissue voxels. The dose to voxel i is given by:

$$d_i = \sum_{n=1}^N \sum_{b=1}^B \sum_{j=2}^J D_{i,n,b} x_{n,b}^j \quad \forall i \in T \cup O$$

To ensure dosage is met to the tumour:

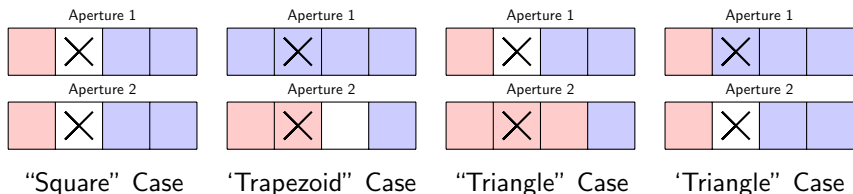
$$\hat{d}_i \leq d_i \quad \forall i \in T$$

Beam on Time Calculations (BOT)

- ▶ We call $x_{n,b}^j$ the BOT which represents the intensity of a beamlet
- ▶ BOT is calculated as:

(dose rate, δ) $\times$ how long the beamlet has been exposed

- ▶ Different leaf motions invoke different BOT calculations



Beam on Time Calculations - Triangle Case

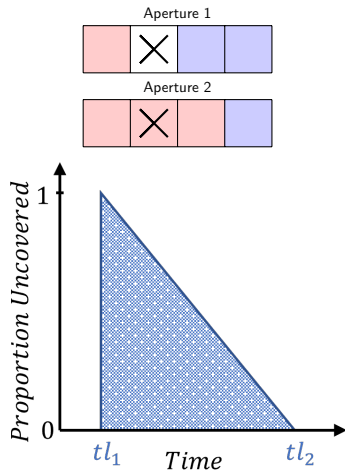
To calculate the BOT, we require the following:

tl_1 = time left leaf arrives
at left boundary of "X"

tl_2 = time left leaf arrives
at right boundary of "X"

Thus:

$$x_{n,b}^j = \delta \times \frac{1}{2} \left(\frac{t}{L_n^j} \right)$$



Beam on Time Calculations

To formulate the if condition, we use the variable $T_{1,n,b}^j \in \{0, 1\}$ such that:

$$T_{L,n,b}^j \geq (1 - \mu_{n,b}^j) + a_{n,b}^{j-1} - 1$$

$$T_{L,n,b}^j \leq (1 - \mu_{n,b}^j)$$

$$T_{L,n,b}^j \leq a_{n,b}^{j-1}$$

which invokes the BOT calculation:

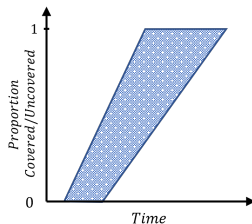
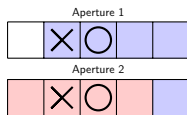
$$x_{n,b}^j - \left(\frac{\delta}{2}\right) \frac{t}{L_n^j} \geq M(T_{L,n,b}^j - 1)$$

$$-x_{n,b}^j + \left(\frac{\delta}{2}\right) \frac{t}{L_n^j} \geq M(T_{L,n,b}^j - 1)$$

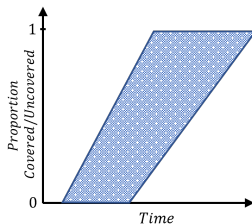
Beam on Time Calculations

The other cases follow similarly. However, the trapezoid case requires special attention.

- Different beamlet positions yield different trapezoids, between the same two apertures



BOT for "X"



BOT for "O"

Beam on Time Calculations

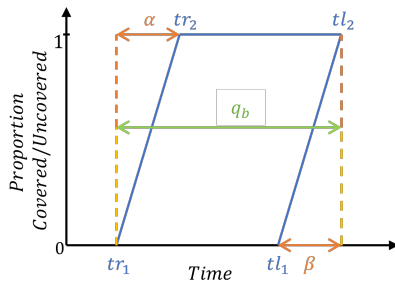
We calculate a general formula, by finding the quantities:

$$tl_2 = t \left(\frac{b - (B - \sum_{k=1}^B l_{n,k}^{j-1})}{L_n^j} \right)$$

$$tr_1 = t \left(\frac{(b-1) - \sum_{k=1}^B r_{n,k}^{j-1}}{R_n^j} \right)$$

$$\alpha = tl_2 - t_1 = \frac{t}{L_n^j}$$

$$\beta = tr_2 - tr_1 = \frac{t}{R_n^j}$$



Beam on Time Calculations

So the required quantity is found to be:

$$q_b = t \left(\frac{b - (B - \sum_{k=1}^B l_{n,k}^{j-1})}{L_n^j} - \frac{(b-1) - \sum_{k=1}^B r_{n,k}^{j-1}}{R_n^j} \right)$$

Then using the formula of a trapezoid:

$$x_{n,b}^j = \frac{\delta}{2} [(q_b - \beta) + (q_b - \alpha)]$$

We complete this constraint by modelling the if conditions, which follow similarly to the previous cases.

Final Model

Let C_O and C_T denote parameters indicating the relative importance of healthy tissue and tumour voxels. The final model follows:

$$\text{Minimize} \quad C_O \sum_{i \in O} d_i + C_T \sum_{i \in T} d_i$$

subject to

Aperture modelling constraints : Slides (12) – (13)

Sliding Motion constraints : Slides (14)

Dose constraints : Slides (15)

Intensity constraints : Slides (16) – (21)

Problem Complexity

The proposed model is very expensive to solve, due to the:

- ▶ non-linear constraints in the BOT calculations
- ▶ use of integer variables
- ▶ sheer size of the problem
 - ▶ approximately 360 000 variables and 1 300 000 constraints

Outline

Introduction

Model Formulation

Heuristic Solution Method

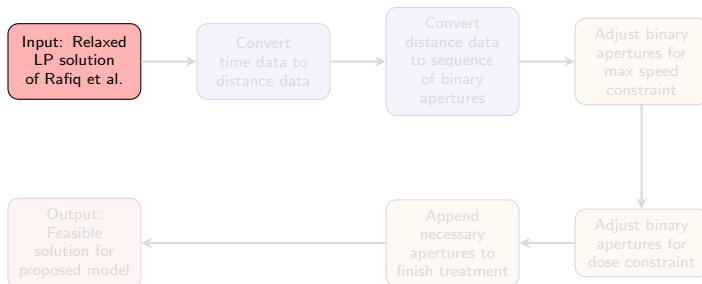
Results

Conclusion

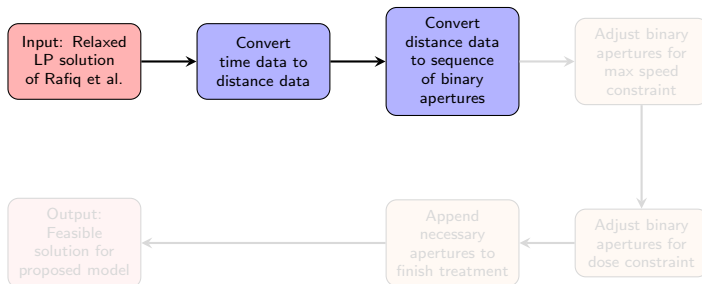
Motivation for Heuristic

- ▶ Generate a close to optimal solution to warm start the model
- ▶ Use for benchmarking the mathematical model (quantifying optimality gap)
- ▶ Starting point for direct clinical implementation of solution

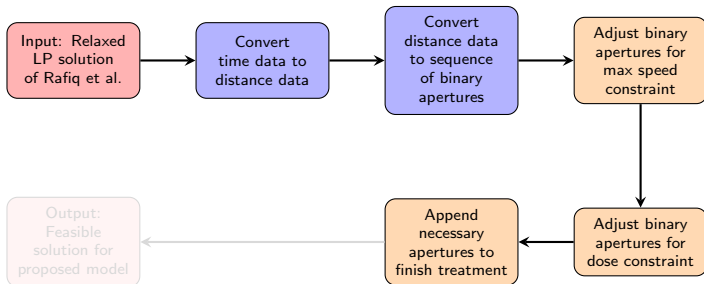
Heuristic Method Overview



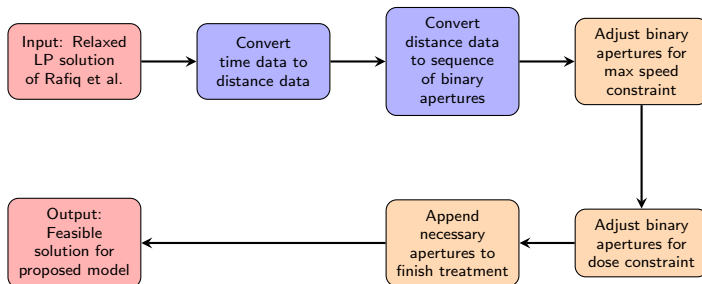
Heuristic Method Overview



Heuristic Method Overview



Heuristic Method Overview



Binary Conversion

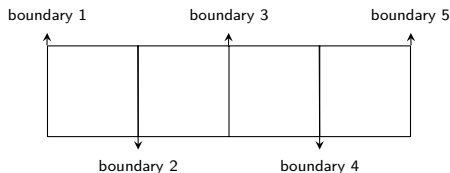


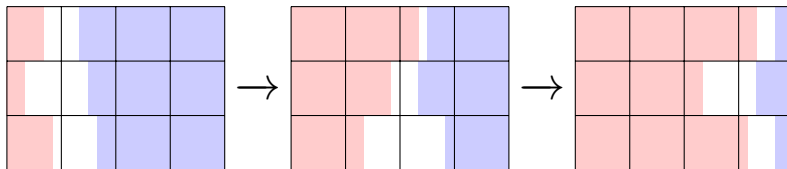
Figure: Beamlets boundaries of some row n

For each boundary in our discretized beam we get four values:

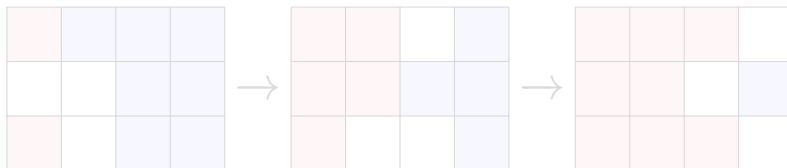
- ▶ time the left leaf arrives and departs a particular boundary
- ▶ time the right leaf arrives and departs a particular boundary

We convert this data to positional data, or apertures at times $0, t, 2t, \dots$

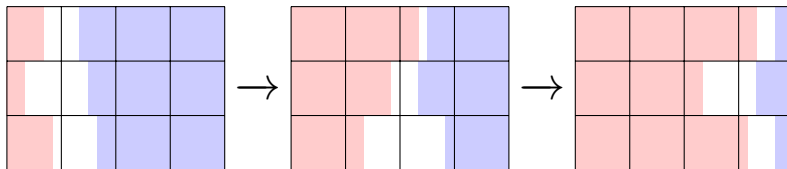
Binary Conversion



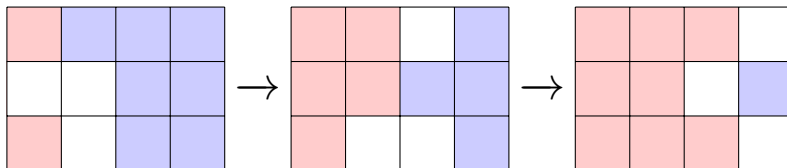
now after rounding to the nearest boundary...



Binary Conversion

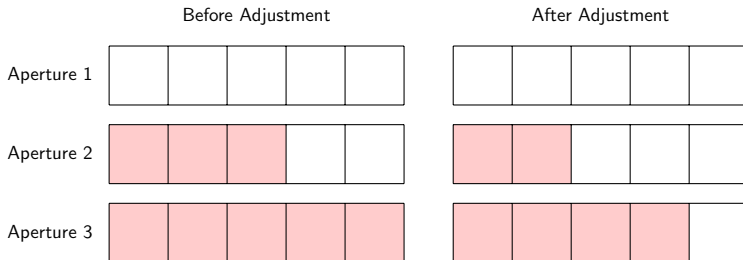


now after rounding to the nearest boundary...



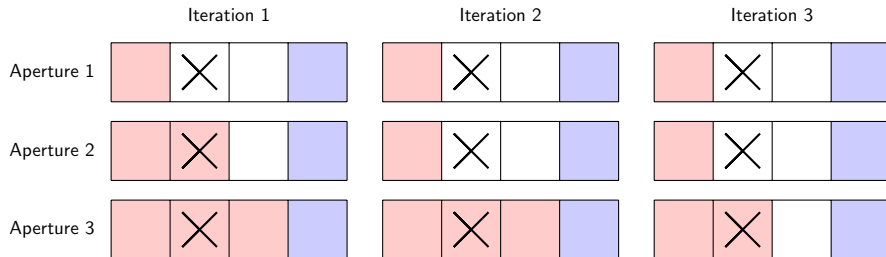
Feasibility Maintenance - Max Speed

The binary approximation may violate our constraints, so we must adjust for them individually. In this example, let $Y = 2$.



Feasibility Maintenance - Dose Constraint

The binary approximation may also introduce beamlets that do not get enough BOT. We hold the left leaves back to adjust for this.



Outline

Introduction

Model Formulation

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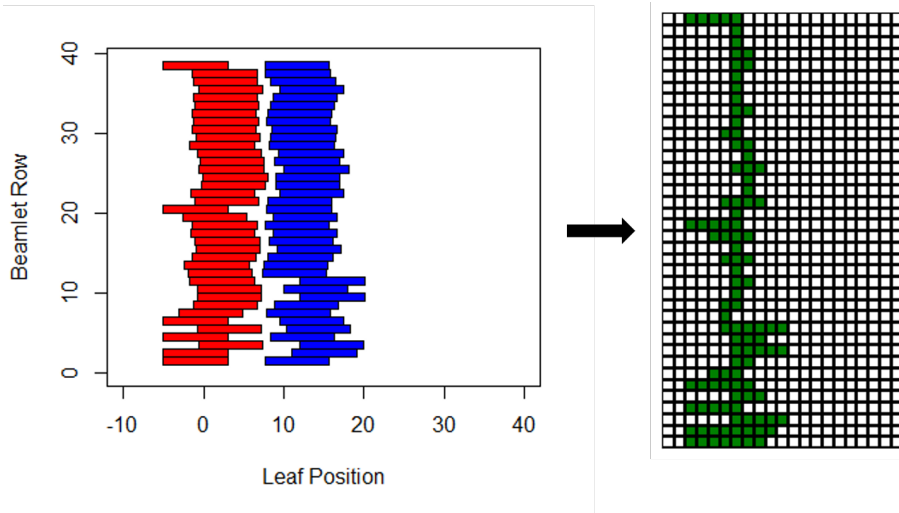
Results

Conclusion

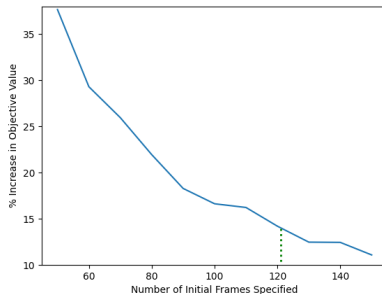
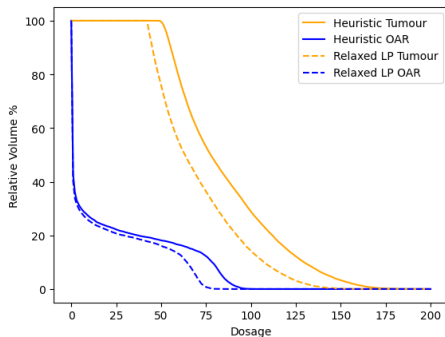
Data and Parameter Specification

- ▶ Breast cancer patient from Grand River Regional Cancer Centre
- ▶ Prescribed dose to tumours: 42.55 Gy, dose rate of 10 MU/s
- ▶ Discretized into 38×24 beam-field, 121 apertures, 1 second between each aperture
- ▶ 18605 tumour voxels, 8267 healthy tissue voxels
- ▶ Max leaf travel: 2 beamlets
- ▶ Equally weighted tumour and healthy tissue importance

Binary Conversion Results



Treatment Quality



Outline

Introduction

Model Formulation

Heuristic Solution Method

Results

Conclusion

Summary

- ▶ Formulated non-linear MIP model which is computationally expensive to solve
- ▶ Proposed a heuristic to generate a feasible solution
- ▶ Promising due to how close to optimal it is

Future Research Directions

- ▶ Experiment with different types of patient data and parameter choices
- ▶ Solve a small dummy problem to validate and compare with heuristic
- ▶ Linearize the model, generalize constraints, and consider relaxations to make the model to make easier to solve

Thank you



Appendix

Appendix I

- Brahme, A., Roos, J. .-, & Lax, I. (1982). Solution of an integral equation encountered in rotation therapy. *Physics in Medicine and Biology*, 27(10), 1221–1229.
<https://doi.org/10.1088/0031-9155/27/10/002>
- Convery, D., & Rosenbloom, M. (1992). The generation of intensity-modulated fields for conformal radiotherapy by dynamic collimation. *Physics in medicine & biology*, 37(6), 1359.
- Craft, D., & Halabi, T. (2018). Sliding window imrt and vmat optimization. In *Decision analytics and optimization in disease prevention and treatment* (pp. 307–322). John Wiley & Sons, Ltd.
<https://doi.org/https://doi.org/10.1002/9781118960158.ch14>

Appendix II

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- Papp, D., & Unkelbach, J. (2014). Direct leaf trajectory optimization for volumetric modulated arc therapy planning with sliding window delivery. *Medical physics*, 41(1), 011701.